Data Engineering [D.E]
Assignment-05

30/30

Name: A. Aswini
Branch:- CSD
Roll_no: 25NUS44401

1) What is a Data model? Illustrate in detail the three main data models with suitable examples.

Data Model:

A Data model represents how data is structured and related to the real world. It defines data elements, relationships and rules to ensure consistency and support business processes. A good data model helps in accurate decision-making, while a poor model leads to confusion

Types of Data models:

1) Conceptual Data model: Represents high-level business logic & rules.
- Describes what dat is needed, not how it is stored
- Focuses on entities & relationships.

example: • Entities: Customer, order
- Relationship: A customer places orders
- No technical details like data types or keys.

2) Logical Data model: Adds more detail to the conceptual model.
- Describes how the data will be structured logically.

example:
- customer table: customer_ID, Name, Address
- order table: order_ID, customer_ID
- Relationship: customer_ID links both tables.

3) Physical Data model:

- Represents actual implementation in a database system.
- Defines how data is stored physically.

Examples:

- Table created in MySQL / PostgreSQL
- Data types like INT, VARCHAR
- Indexes on Customer_ID for faster queries.

Conceptual → logical → physical

fig: The continuum of data models: conceptual, logical & physical

- Conceptual → Abstract view
- Logical → Detailed structure
- physical → Actual implementation.

2) How does embedded analytics empower organizations to integrate data insights within applications.

Embedded analytics refers to the integration of analytical capabilities directly into applications, allowing users to access data insights within the system they are using.

It empowers organizations by:

- providing real-time or near real-time insights inside applications
- Enabling faster decision-making without switching to separate tools
- offers user-friendly dashboards for both technical and non-technical users.
- supporting data-driven actions such as adjusting operations or strategies instantly.

Cross Join: when each row of first table is combined with every row from the second table is known as Cross Join.
(cartesian product (all combination)

ensuring high performance with low latency, fast queries and high concurrency.

Example:
A Smart thermostat app displays real-time temperature and energy usage, helping users optimize energy consumption. Similiarly e-commerce dashboards show live sales and inventory data

Embedded analytics transforms applications into data-driven systems by delivering insights directly to users, enabling real-time decisions, improved efficiency and better user engagement.

3) Briefly explain the different ways to serve data for analytics and ML-File exchange databases, Query federation and serving notebooks.

1) File exchange: Data is shared as files between systems or teams. common formats: CSV, JSON, Parquet.
Features: • Simple & widely used
• Suitable for batching processing
• Easy to transfer via cloud storage or FTP
Example: sharing daily sales data as a CSV file
Limitation: Not real time.
• version control & consistency issues

2) Databases: Data is stored & served through databases
• Includes: Relational databases, NoSQL databases.
Features: Supports structured querying (SQL)
• provides fast access & updates.
• Good for both analytics & ML pipelines.
Example: Using a database to retrieve customer data for analysis
Limitations: Scaling can be complex for very large data.

3) Query Federation: Query data from multiple sources without moving it

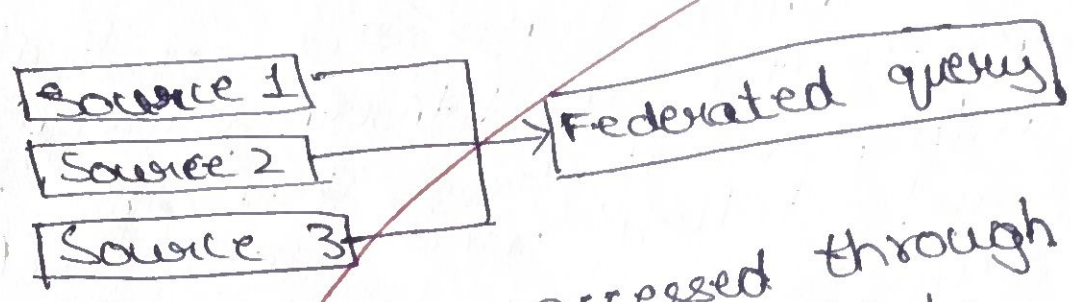
- combines data easily

Features: Reduces data duplication

- Useful for distributed environments & data bases together

example: Querying data from cloud storage

Limitations: Can have performance issues due to distributed queries.

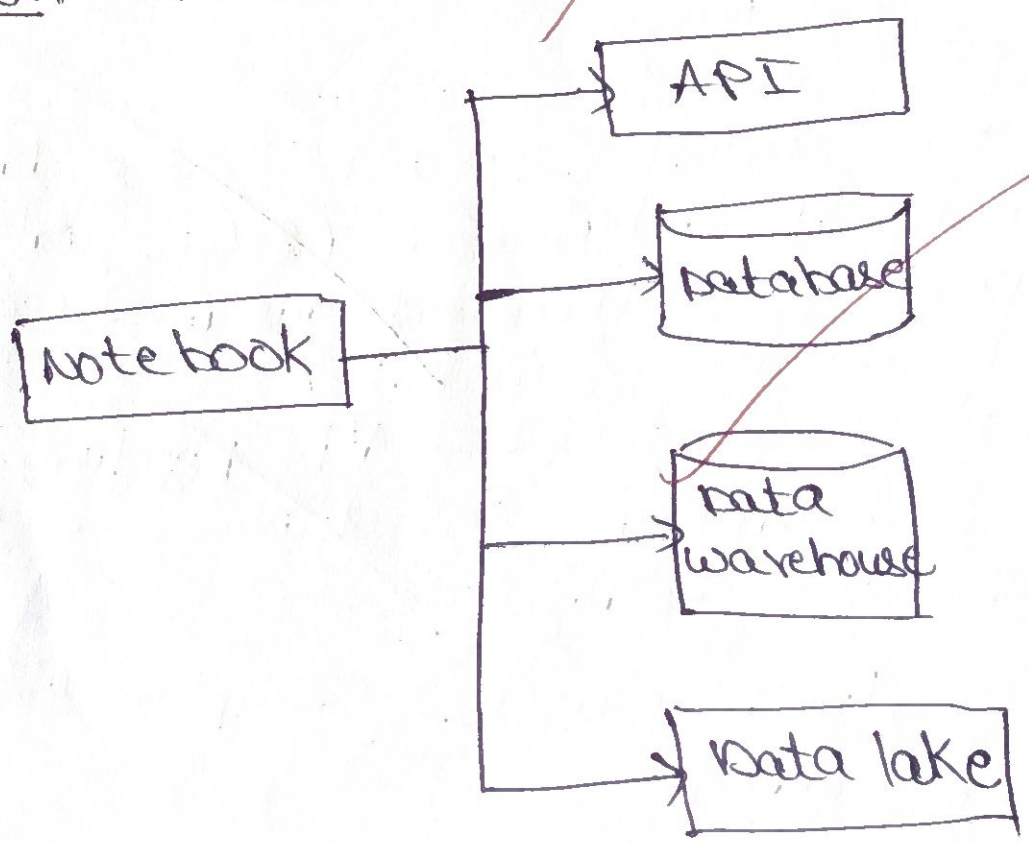


4) Serving Notebooks: Data is accessed through notebooks used by analysts and data scientists.

example :- Jupyter notebooks.

Features: Interactive data exploration

widely used in ML work flows

example: Data scientist analyzing data & training ML models in a notebook.

Limitation: Limited scalability